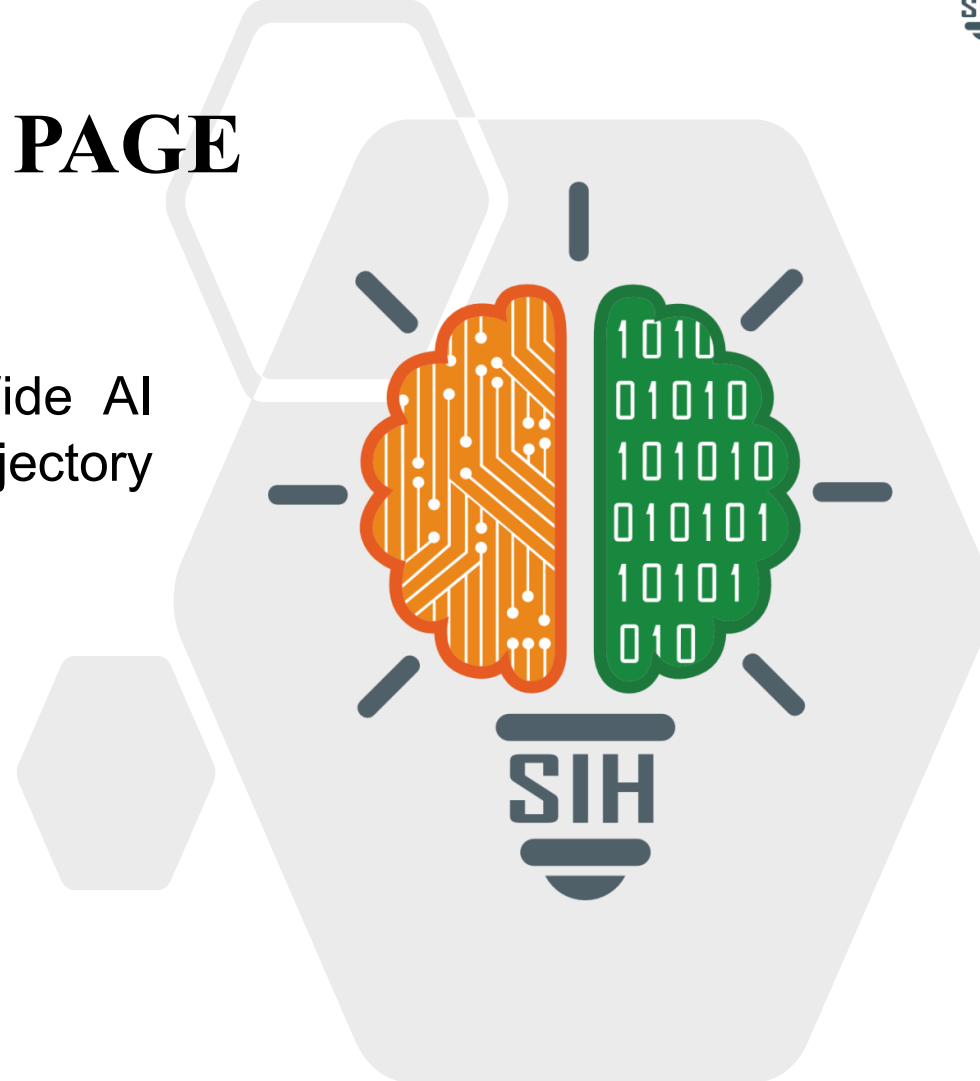


TITLE PAGE

- **Problem Statement ID** – 26127
- **Problem Statement Title-** City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**143680
- **Team Name** - NeuraTrack



1 The Problem

Why existing CCTV tracking fails



Blurred / Unreadable Number Plates

Low resolution, motion blur, dirty or damaged plates.



Adverse Conditions

Night, rain, glare, lighting changes.



Partial Occlusion

Vehicles blocked by other vehicles, buses, trucks or pedestrians.



Different Viewpoints

Same vehicle looks different across cameras (front, back, side).



Missing Observations

Camera outages, network issues, maintenance periods.



False Visual Similarity

Different vehicles may look similar (same make, colour, model).



A single CCTV detection is an observation, not a trustworthy vehicle identity.

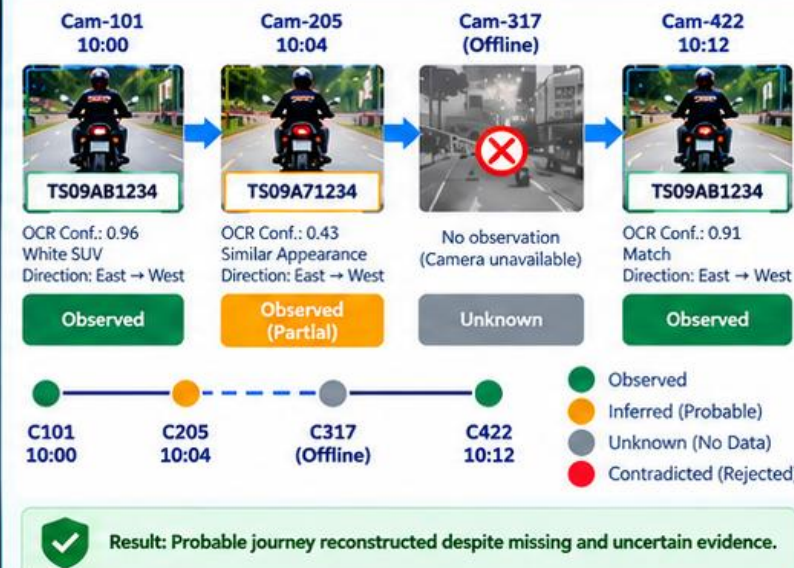
2 Our Proposed Solution — TRACE-X

An agentic AI system that reasons, adapts and reconstructs vehicle journeys from imperfect evidence



3 Example: Reconstructing a Journey

Connecting multiple cameras to build a probable journey



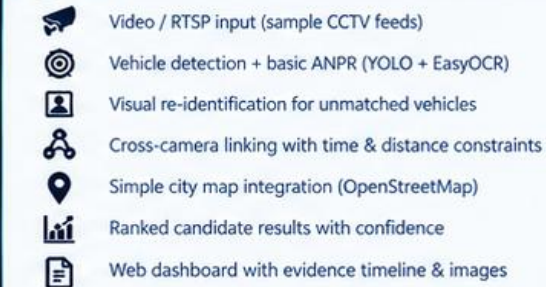
4 How TRACE-X Addresses the Problem

Real-World Challenge	TRACE-X Response
Blurred / unreadable plates	Visual Re-ID + image enhancement + confidence scoring
Adverse conditions (night, rain, glare)	Quality assessment + robust models trained on low-quality data
Partial occlusion	Multi-camera fusion + appearance matching
Different viewpoints	Cross-camera re-identification (front, back, side)
Missing / offline cameras	Handles unknown/missing observations, fills gaps using route & time
Similar-looking vehicles	Evidence fusion + contradiction detection
Multiple possible routes	Agentic reasoning with road network constraints
Uncertain results	Confidence estimation + explainable evidence graph + human review

5 Innovation & Uniqueness

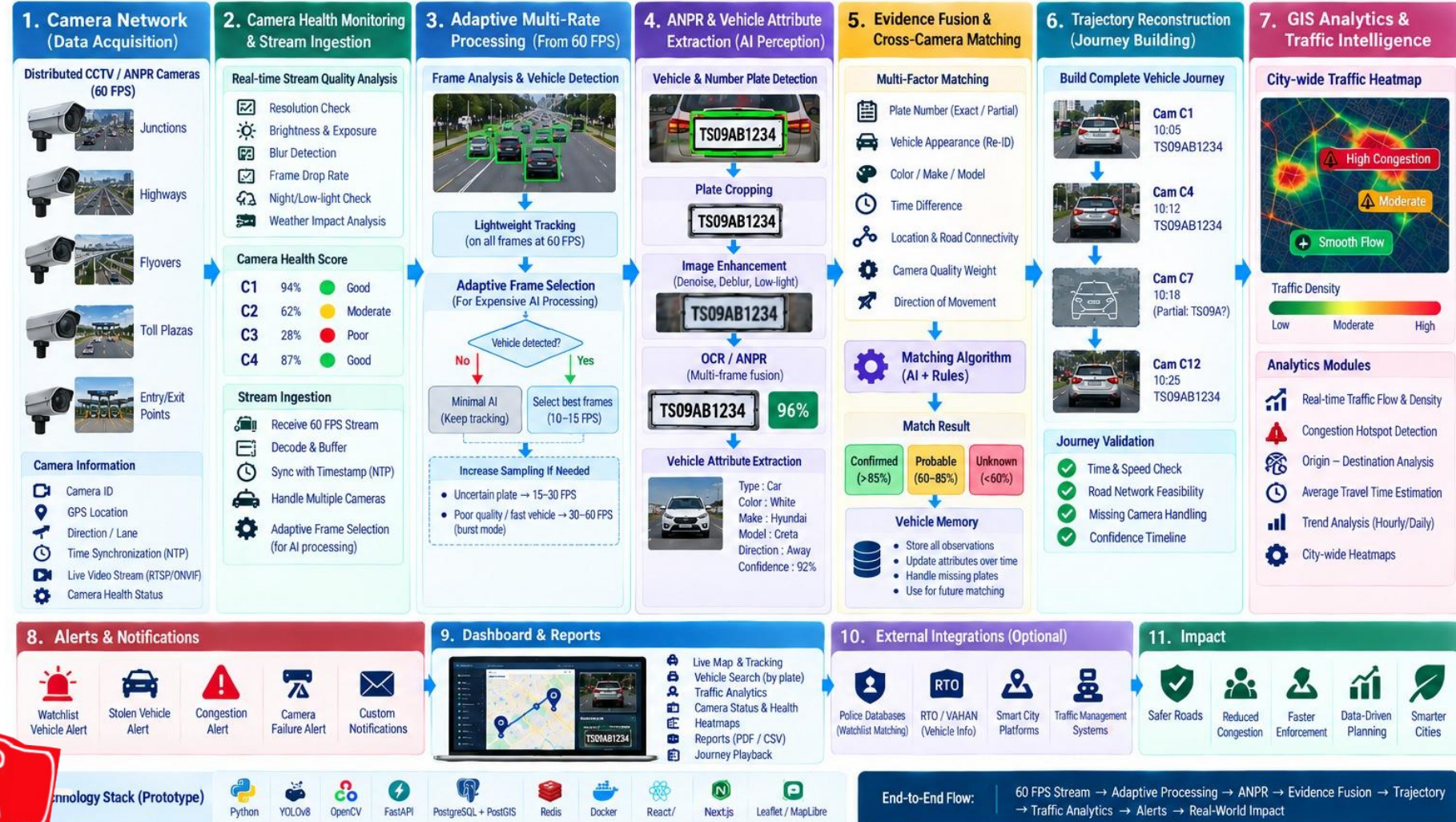
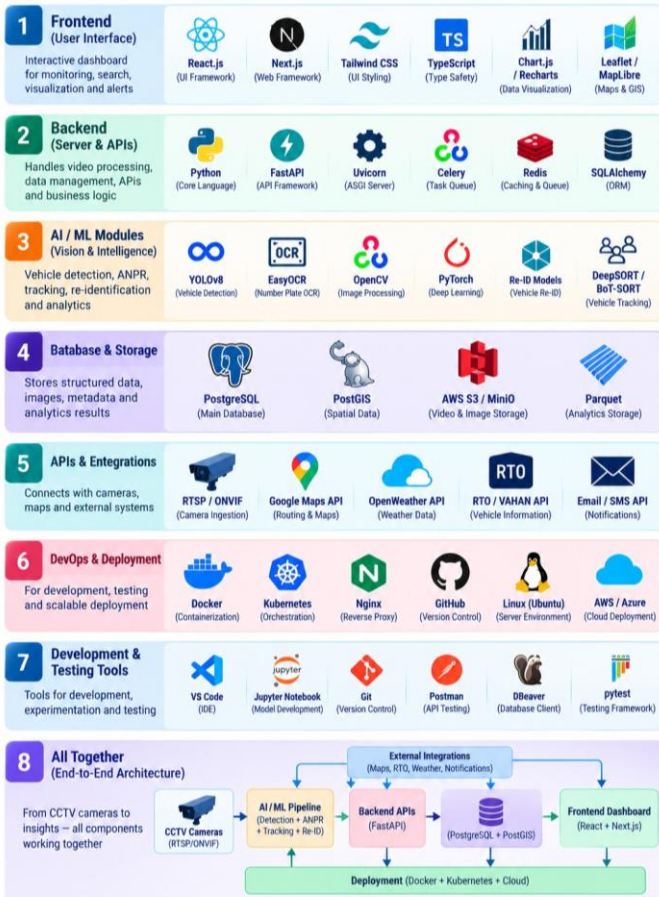


6 Prototype (MVP Features)



Neura Track

TECHNICAL APPROACH



video



FEASIBILITY ANALYSIS



 **Infrastructure Reusability:**
Builds on existing CCTV/ANPR camera networks — no new hardware installation required.

 **Scalable Deployment:**
Pilot at a few key intersections first, then expand citywide zone-by-zone.

 **Cost-Effectiveness:**
Open-source CV stack keeps setup cost low versus proprietary ANPR systems.

 **Authority Integration:**
Blacklist alerts, case management, and manual override for police control.

 **Data Readiness:**
Police FIR records and camera metadata plug directly into the registry and blacklist.

VIABILITY



 **Proven Concept:**
Mirrors AI traffic-surveillance systems already live in major Indian cities.

 **Builds Public Trust:**
Automated, timestamped detection minimizes bias and gives auditable evidence.

 **Authority Engagement:**
Real-time dashboards keep control-room operators actively in the loop.

 **Guarantees Accountability:**
Every detection, alert, and case action is logged and traceable.

 **Regulatory Alignment:**
Fits existing FIR-linked enforcement and investigation workflows, no new legislation needed.

BUSINESS POTENTIAL



 **Government Cost Savings:**
Automates manual monitoring, cutting personnel hours and cost.

 **New Revenue Generation:**
OCR output integrates with e-challan systems for automated fines.

 **Faster Vehicle Recovery:**
Cross-camera trajectory search cuts investigation to minutes.

 **Multi-City Scalability:**
Same architecture licenses to other cities with local config only.

 **Analytics as a Product:**
Aggregated data supports signal timing and infra planning.



CURRENT SITUATION
Fragmented. Delayed. Uncertain.

- Time-consuming Investigations**
Manual search across multiple cameras
- Incomplete / Blurred Evidence**
Missing or unclear vehicle data
- Missed Connections**
No easy way to link cameras across the city
- Higher Congestion**
Limited real-time insights
- Increased Costs**
More manpower, more time, more fuel waste

WITH TRACE-X
Faster. Smarter. More Reliable.

- Faster Investigations**
Trace vehicle journeys in minutes
- Verified Evidence**
AI + multi-camera correlation
- Connected City View**
Links cameras across all directions (N, S, E, W)
- Data-Driven Traffic Management**
Real-time insights & heatmaps
- Lower Social & Environmental Cost**
Less congestion, lower emissions



REAL IMPACT. BETTER TOMORROW.

- Faster Response**
Helps trace vehicles quickly for accidents, crime and emergencies.
- Safer Roads**
Identifies suspicious and stolen vehicles, improving public safety.
- Reduced Congestion**
Insights from vehicle movement help in better traffic planning.
- Cleaner Environment**
Smoother traffic flow reduces fuel wastage and emissions.
- More Accountable Cities**
Data-driven decisions and evidence-based enforcement.
- Economic Benefits**
Saves time, fuel and resources, boosting city productivity.

Key Benefits of Trace X

- Faster Investigations**
Quickly trace vehicle movement across multiple cameras.
- Improved Traffic Management**
Real-time insights for smoother and safer traffic flow.
- Proactive Alerts**
Detect stolen, wanted or suspicious vehicles instantly.
- Data-Driven Decisions**
Accurate analytics for better urban planning.
- Increased Public Safety**
Helps law enforcement and creates safer cities.

Reduction in Investigation Time
(Average time to trace a vehicle)



System	Time (hours)
Without Trace X	36 hrs
With Trace X	4 hrs

Helps law enforcement find vehicles in minutes instead of hours.

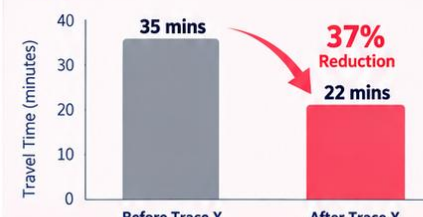
Increase in Vehicle Detection Accuracy
(Correctly identified vehicles)



System	Accuracy (%)
Traditional Systems	65%
With Trace X	95%

More accurate detection even in challenging conditions (night, rain, crowded traffic).

Reduction in Traffic Congestion
(Average travel time on monitored roads)



System	Travel Time (minutes)
Before Trace X	35 mins
After Trace X	22 mins

Smarter traffic management through real-time vehicle analytics.

Increase in Detection of Stolen/Wanted Vehicles
(Monthly detections)



System	Number of Detections
Before Trace X	50
With Trace X	180

Helps recover stolen vehicles and identify wanted vehicles faster.

WIDER IMPACT

- Citizens**
Safer, smoother, stress-free travel
- Authorities**
Faster action, better control
- Environment**
Cleaner, greener cities
- Economy**
Higher productivity, lower costs



Click For More info





1 RESEARCH BASE

[Indian ANPR • 2022](#)

Indian plate diversity makes direct transfer from non-Indian datasets difficult.

[Multi-Camera Tracking • 2019](#)

Visual + orientation + spatial-temporal cues improve vehicle tracking across cameras.

[City-Scale MCT • 2021](#)

Re-ID + track matching + spatial-temporal information support city-scale traffic management.

[Intelligent Camera Selection • 2022](#)

Camera-selection policies can reduce unnecessary Re-ID queries and computation.

3 VALIDATION FOUNDATION

[Hyderabad CCTV](#)

IITH_Helmet2 / IITH_Accident

[City-scale benchmark](#)

CityFlow / AI City

[Indian ANPR](#)

Indian plate dataset + BARC study

[Road network](#)

OpenStreetMap for GIS topology

2 TRACE-X RESEARCH EXTENSION

1

A. Confidence-aware evidence

Every observation carries OCR, image-quality and camera-health confidence.

2

B. Multi-hypothesis journey

Keep competing routes instead of forcing one vehicle path when evidence is ambiguous.

3

C. Negative + contradictory evidence

Represent expected-but-unseen observations and reject physically inconsistent transitions.

4

D. Evidence-driven camera selection

Choose the next camera/time-window by expected information gain, not only proximity.

Links

